forming, perhaps, the backdrop for other occurrences. These two ways of looking at death (a popular, if morbid, example) are illustrated by:

> When Queen Anne died, the Whigs brought in George.
> While Queen Anne was dying, the Jacobites hatched treasonable plots.

Another feature of linguistic interest is the peculiar nature of *accomplishment* verbs, illustrated by:

(1) The Amalgated Conglomerate Building was built during the period March–August, 1972.
(1′) The ACB was built during the period April–July, 1972.
(2) The ACB was being built (i.e. was under construction) during the period March–August, 1972.
(2′) The ACB was under construction during the period April–July, 1972.

Note that (1) and (1′) are inconsistent, whereas (2) implies (2′)!

In part, the switch is motivated by a philosophical belief that periods are somehow more basic than instants. This motivation would be more convincing were 'periods' not assumed (as they are in too many recent works) to have sharply-defined (i.e. *instantaneous*) beginnings and ends. It may also be remarked that at the level of *experience* some occurrences do *appear* to be instantaneous (i.e. we don't *discern* stages in them). Thus 'bubbles when they burst' *seem* to do so 'all at once and nothing first'. While at the level of *reality*, some occurrences of the sort studied in quantum physics may well take place instantaneously, just as some elementary particles may well be pointlike. Thus the philosophical belief that every occurrence takes some time (period) to occur is not *obviously* true on any level.

Now for the mechanics of the switch: For any frame (X, R) we consider the set $I(X, R)$ of nonempty bounded open intervals of form $\{z : xRz \wedge zRy\}$. Among the many relations on this set that could be defined in terms of R we single out two:

$$\textit{Inclusion}: \ a \subseteq b \quad \text{iff} \quad \forall x(x \in a \rightarrow x \in b),$$
$$\textit{Order}: \ a \lhd b \quad \text{iff} \quad \forall x \forall y(x \in a \wedge y \in b \rightarrow xRy).$$

To any class $\mathscr{K}$ of frames we associate the class $\mathscr{K}'$ of those structures of form $(I(X, R), \subseteq, \lhd)$ with $(X, R) \in \mathscr{K}$, and the class $\mathscr{K}^+$ of those structures (Y, S, T) that are isomorphic to elements of $\mathscr{K}'$.

A first problem in switching from instants to periods as the basis for the logic of time is to find each important class $\mathscr{K}$ of frames a set of postulates whose models will be precisely the structures in $\mathscr{K}^+$. For the case of dense total orders without extrema, and for some other cases, suitable postulate sets are known, though none is very elegant. Of course this first problem is not yet a problem of *tense* logic; it belongs rather to applied first- and second-order logic.

To develop a period-based *tense* logic we define a *valuation* in a structure (Y, S, T) – where S, T are binary relations on Y – to be a function V assigning each p_i a subset of Y. Then from among all possible connectives that could be defined in terms of S and T, we single out the following:

$$
\begin{aligned}
V(\neg\alpha) &= Y - V(\alpha)\\
V(\alpha \wedge \beta) &= V(\alpha) \cap V(\beta)\\
V(\nabla\alpha) &= \{a : \forall b(bSa \to b \in V(\alpha))\}\\
V(\Delta\alpha) &= \{a : \forall b(aSb \to b \in V(\alpha))\}\\
V(F\alpha) &= \{a : \exists b(aTb \wedge b \in V(\alpha))\}\\
V(P\alpha) &= \{a : \exists b(bTa \wedge b \in V(\alpha))\}.
\end{aligned}
$$

The main *technical* problem now is, given a class $\mathbf{L}$ of structures (Y, S, T) – for instance, one of form $\mathbf{L} = \mathscr{K}^+$ for some class $\mathscr{K}$ of frames – to find a sound and complete axiomatization for the tense logic of $\mathbf{L}$ based on the above connectives. Some results along these lines have been obtained, but none as definitive as those of instant-based tense logic reported in Section 2. Indeed, the choice of relations ($\subseteq$ and $\lhd$), and of admissible classes $\mathbf{L}$ (should we only consider classes of form $\mathscr{K}^+$?), and of conncectives ($\neg, \wedge, \Delta, \nabla, F, P$), and of admissible valuations (should we impose restrictions, such as requiring $b \in V(p_i)$ whenever $a \in V(p_i)$ and $b \subseteq a$?) are all matters of controversy.

The main problem of *interpretation* – one to which advocates of period-based tense logic have perhaps not devoted sufficient attention – is how to make intuitive sense of the notion $a \in V(p)$ of a sentence p being true *with respect to* a time-period a. One proposal is to take this as meaning that p is true *throughout* a. Now given a valuation W in a frame (X, R), we can define a valuation $I(W)$ in $I(X, R)$ by $I(W)(p_i) = \{a : a \subseteq W(p_i)\}$. When and *only* when V has the form $I(W)$ is 'p is true throughout a' a tenable reading of $a \in V(p)$. It is not, however, easy to characterize intrinsically those V that admit a representation in the form $V = I(W)$. Note that even in this case, $a \in V(\neg p)$ does not express '$(\neg p)$ is true throughout a' (but rather '$\neg(p$ is true throughout $a)$'). Nor does $a \in V(p \vee q)$ express '$(p \vee q)$ is true throughout a'.

Another proposal, originating in Burgess [1982] is to read $a \in V(p)$ as

'‡ is *almost* always true during a'. This reading is tenable when V has the form $J(W)$ for some valuation W in (X, R), where $J(W)(p_i)$ is by definition $\{a : a - W(p_i)$ is nowhere dense in the order topology on $(X, R)\}$. In this case, '$(\neg p)$ is almost always true during a' is expressible by $a \in V(\nabla\neg p)$, and '$(p \vee q)$ is almost always true during a' by $a \in V(\nabla\neg\nabla\neg(p \vee q))$. But the whole problem of interpretation for period-based tense logic deserves more careful thought.

There have been several proposals to redo tense logic on the basis of 3- or 4- of multi-valued truth-functional logic. It is tempting, for instance, to introduce a truth-value 'unstatable' to apply to, say, 'Bertrand Russell is smiling' in 1789. In connection with the switch from instants to periods, some have proposed introducing new truth-values 'changing from true to false' and 'changing from false to true' to apply to, say, 'the rocket is at rest' at take-off and landing times. Such proposals, along with proposals to combine, say, tense logic and intuitionistic logic, lie beyond the scope of this survey.

6. GLIMPSES AROUND

6.1. *Metric Tense Logic*

In *metric* tense logic we assume Time has the structure of an ordered Abelian group. We introduce variables $x, y, z, \ldots$ ranging over group elements, and simples $0, +, <$ for the group identity, addition, and order. We introduce operators $\mathscr{F}$. $\mathscr{P}$ joining terms for group elements with formulas. Here, for instance, $\mathscr{F}(x + y)(p \wedge q)$ means that it will be the case $(x + y)$ time-units hence that p and q. Metric tense logic is intended to reflect such ordinary-language *quantitative* expressions as '100 years from now' or 'tomorrow about this time' or 'in less than five minutes'. The *qualitative* F, P of nonmetric tense logic can be recovered by the definitions $Fp \longleftrightarrow \exists x > 0 \mathscr{F}xp$, $Pp \longleftrightarrow \exists x > 0 \mathscr{P}xp$. Actually, the 'ago' operator $\mathscr{P}$ is definable in terms of the 'hence' operator $\mathscr{F}$ since $\mathscr{P}xp$ is equivalent to $\mathscr{F}\text{-}xp$. It is not hard to write down axioms for metric tense logic whose completeness can be proved by a Henkin-style argument.

But decidability is lost: The decision problem for metric tense logic is easily seen to be equivalent to that for the set of all universal monadic (second-order) formulas true in all ordered Abelian groups. We will show that the decision problem for the validity of first-order formulas involving a single two-place predicate $\in$ – which is well known to be unsolvable – is reducible to the latter: Given a first-order $\in$-formula φ, fix two one-place predicate variables U, V. Let φ_0 be the result of restricting all quantifiers in φ

to U (i.e. $\forall x$ is replaced by $\forall x(U(x) \rightarrow \ldots)$ and $\exists x$ by $\exists x(U(x) \wedge \ldots)$.) Let φ_1 be the result of replacing each atomic subformula $x \in y$ of φ_0 by $\exists z(V(z) \wedge V(z+x) \wedge V(z+x+y))$. Let φ_2 be the universal monadic formula $\forall U \forall V(\exists x U(x) \rightarrow \varphi_1)$. Clearly if φ is logically valid, then so is φ_2 and, in particular, the latter is true in all ordered Abelian groups. If φ is not logically valid, it has a countermodel consisting of the positive integers equipped with a binary relation E. Consider the product $\mathbb{Z} \times \mathbb{Z}$ where $\mathbb{Z}$ is the additive group of integers; addition in this group is defined by $(x, y) + (x', y') = (x + x', y + y')$; the group is orderable by $(x, y) < (x', y')$ iff $x < x'$ or ($x = x'$ and $y < y'$). Interpret U in this group as $\{(n, 0): n > 0\}$; interpret V as the set consisting of the $(2^m 3^n, 0)$, $(2^m 3^n, m)$, and $(2^m 3^n, m+n)$ for those pairs (m, n) with mEn. This gives a countermodel to the truth of φ_2 in $\mathbb{Z} \times \mathbb{Z}$. Thus the desired reduction of decision problems has been effected.

Metric tense logic is, in a sense, a hybrid between the 'regimentation' and 'autonomous tense logic' approaches to the logic of time. Other hybrids of a different sort – not easy to describe briefly – are treated in an interesting paper of Bull [1970].

6.2. *Time and Modality*

As mentioned in the introduction, Prior attempted to apply tense logic to the exegesis of the writings of ancient and mediaeval philosophers and logicians (and for that matter of modern ones such as C. S. Peirce and J. Łukasiewicz) on future contingents. The relations between tense and mood or modality is properly the topic of Richmond H. Thomason's chapter in this volume (II.3).

We can, however, briefly consider here the topic of so-called *Diodorean* and *Aristotelian* modal fragments of a tense logic L. The former is the set of modal formulas that become theses of L when $\Box p$ is defined as $p \wedge Gp$; the latter is the set of modal formulas that become theses of L when $\Box p$ is defined as $Hp \wedge p \wedge Gp$. Though these seem far-fetched definitions of 'necessity', the attempt to isolate the modal fragments of various tense logics undeniable was an important stimulus for the earlier development of our subject. Briefly the results obtained can be tabulated as follows. It will be seen that the modal fragments are usually well-known C. I. Lewis systems.

Class of frames	Tense logic	Diodorean fragment	Aristotelian fragment
All frames	$\mathbf{L}_0$	T (= M)	B
Partial orders	$\mathbf{L}_1$	S4	B
Lattices	$\mathbf{L}_9$	S4.2	B
Total orders Dense orders	$\mathbf{L}_2$, $\mathbf{L}_5$	S4.3	S5

The Diodorean fragment of the tense logic $\mathbf{L}_6$ of discrete orders has been determined by M. Dummett; the Aristotelian fragment of the tense logic of trees has been determined by G. Kessler. See also our comments below on R. Goldblatt's work.

6.3. *Relativistic Tense Logic*

The *cosmic* frame is the set of all point-events of space-time equipped with the relation of *causal accessibility*, which holds between u and v if a signal (material or electromagnetic) could be sent from u to v. The $(n + 1)$-dimensional *Minkowski* frame is the set of $(n + 1)$-tuples of real numbers equipped with the relation which holds between $(a_0, u_1, \ldots, a_n)$ and $(b_0, b_1, \ldots, b_n)$ iff:

$$\sum_{i\ 1}^{n} (b_n - a_n)^2 - (b_0 - a_0)^2 > 0 \quad \text{and} \quad b_0 > a_0.$$

For present purposes, the content of the *special theory of relativity* is that the cosmic frame is isomorphic to the 4-dimensional Minkowski frame.

A little calculating shows that any Minkowski frame is a lattice without maximum or minimum, hence the tense logic of special relativity should at least include $\mathbf{L}_9$. Actually we will also want some axioms to express the density and continuity of a Minkowski frame. A surprising discovery of Goldblatt [1980] is that the *dimension* of a Minkowski frame influences its tense logic. Indeed, he shows that for each n there is a formula γ_{n+1} which is valid in the $(m + 1)$-dimensional Minkowski frame iff $m < n$. For example, writing Ep for $p \wedge Fp$, γ_2 is:

$$\begin{gathered} Ep \wedge Eq \wedge Er \wedge \neg E(p \wedge q) \wedge \neg E(p \wedge r) \wedge \neg E(q \wedge r) \rightarrow \\ E((Ep \wedge Eq) \vee (Ep \wedge Er) \vee (Eq \wedge Er)). \end{gathered}$$

On the other hand, he also shows that the dimension of a Minkowski frame does *not* influence the Diodorean modal fragment of its tense logic: The

Diodorean modal logic of special relativity is the same as that of arbitrary lattices, namely **S4.2**. Combining Goldblatt's argument with the 'trousers world' construction in general relativity, should produce a proof that the Diodorean modal fragment of the latter is the same as that of arbitrary partial orders, namely **S4**.

Despite recent advances, the tense logic of special relativity has not yet been completely worked out; that of general relativity is even less well understood. Burgess [1979] contains a few additional philosophical remarks.

6.4. *Thermodynamic Time*

One of the oldest metaphysical conceits (found in Hindu theology and pre-Socratic philosophy, and in modern psychological dress in Nietzsche and celestial mechanical dress in Poincaré) is that everything that has ever happened is destined to be repeated over and over again. This leads to a degenerate tense logic containing the principles $Gp \to Hp$ and $Gp \to p$ among others.

An antithetical view is that traditionally associated with the Second Law of Thermodynamics, according to which irreversible changes are taking place that will eventually drive the Universe to a state of 'heat-death', after which no further change on a macroscopically observable level will take place. The tense logic of this view, which raises several interesting technical points, has been investigated by S. K. Thomason [1972]. The first thing to note is that the principle:

(A10) $\quad GFp \to FGp$

is acceptable for p expressing propositions about macroscopically observable states of affairs provided these do not contain hidden time references; e.g. p could be 'there is now no life on Earth', but not 'particle κ currently has a momentum of precisely k gram-meters/second' or 'it is now an even number of days since the Heat Death occurred'. For the antecedent of (A10) says that arbitrarily far in the future there will be times when p is the case. But for the p that concern us, the truth-value of p is never supposed to change after the Heat Death. So in that case, there will come a time after which p is always going to be true, in accordance with the consequent of (A10).

The question now arises, how can we *formalize* the restriction of p to a special class of sentences? In general, propositions are represented in the formal semantics of tense logic by subsets of X in a frame (X, R). A restricted class of propositions could thus be represented by a distinguished family $\mathscr{B}$ of subsets of X. This motivates the following definition: An *augmented* frame is a triple $(X, R, \mathscr{B})$ where (X, R) is a frame, $\mathscr{B}$ a subset of the power set $\mathscr{P}(X)$ of X closed under complementation, finite intersection, and the operations:

$$gA = \{x \in X : \forall y \in X(xRy \to y \in A)\}$$
$$hA = \{x \in X : \forall y \in X(yRx \to y \in A)\}.$$

A *valuation* in $(X, R, \mathscr{B})$ is a function V assigning each variable p_i an element of $\mathscr{B}$. The closure conditions on $\mathscr{B}$ guarantee that we will then have $V(\alpha) \in \mathscr{B}$ for *all* formulas α. It is now clear how to define validity. Note that if $\mathscr{B} = \mathscr{P}(X)$, then validity in $(X, R, \mathscr{B})$ reduces to validity in (X, R); otherwise more formulas may be valid in the former than the latter.

It turns out that the extension $\mathbf{L}_{10}$ of $\mathbf{L_Q}$ obtained by adding A10 is (sound and) complete for the class of augmented frames $(X, R, \mathscr{B})$ in which (X, R) is a dense total order without maximum or minimum and:

$$\forall B \in \mathscr{B} \exists x(\forall y(xRy \to y \in B) \vee \forall y(xRy \to y \notin B))$$

We have given complete axiomatizations for many intuitively important classes of frames. We have not yet broached the questions: When does the tense logic of a given class of frames admit a complete axiomatization? When does a given axiomatic system of tense logic correspond to some class of frames in the sense of being complete for that class? For information on these large questions, and for bibliographical references, we refer the reader to Johan van Benthem's chapter in this volume on so-called 'Correspondence Theory' (II.4). Suffice it to say here that positive general theorems are few, counterexamples many. The thermodynamic tense logic $\mathbf{L}_{10}$ exemplifies one sort of pathology. Though it is not inconsistent, there is *no* (unaugmented) frame in which all its theses are valid!

6.5. *Quantified Tense Logic*

The interaction of temporal operators with universal and existential quantifiers raises many difficult issues, both philosophical (over identity through changes, continuity, motion and change, reference to what no longer exists or does not exist, essence, and many, many more) and technical (over undecidability, nonaxiomatizability, undefinability or multi-dimensional operators, and so forth) that it is pointless to attempt even a survey of the subject in a paragraph or two. We therefore refer the reader to Nino Cocchiarella's [II.6] and James W. Garson's [II.5] chapters on this subject in this volume.

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CHAPTER II.3

COMBINATIONS OF TENSE AND MODALITY

by RICHMOND H. THOMASON

1. INTERACTIONS WITH TIME

Physics should have helped us to realize that a temporal theory of a phenomenon X is, in general, more than a simple combination of two components: the statics of X and the ordered set of temporal instants. The case in which all functions from times to world-states are allowed is uninteresting; there are too many such functions, and the theory has not begun until we have begun to restrict them. And often the principles that emerge from the interaction of time with the phenomena seem new and surprising. The most dramatic example of this, perhaps, is the interaction of space with time in relativistic space-time.

The general moral, then, is that we shouldn't expect the theory of time $+ X$ to be obtained by mechanically combining the theory of time and the theory of X.[1]

Probability is a case that is closer to our topic. Much ink has been spilled over the evolution of probabilities: take, for instance, the mathematical theory of Markov processes (Howard [1971a, b] make a good text), or the more philosophical question of rational belief change (see, for example, Chapter 11 of Jeffrey [1965], and Harper [1975].) Again, there is more to these combinations than can be obtained by separate reflection on probability measures and the time axis.

Probability shares many features with modalities and, despite the fact that (classical) probabilities are numbers, perhaps in some sense probability *is* a modality. It is certainly the classic case of the use of possible worlds in

D. Gabbay and F. Guenthner (eds.), Handbook of Philosophical Logic, Vol. II, 135–165.

interpreting a calculus. (Sample points in a state space are merely possible worlds under another name.) But the literature on probability is enormous, and almost none of it is presented from the logician's perspective. So, aside from the references I have given, I will exclude it from this survey. However, it seems that the techniques we will be using can also help to illuminate problems having to do with probability; this is illustrated by papers such as D. Lewis [1981] and Van Fraassen [1971a]. For lack of space, these are not discussed in the present essay.

2. INTRODUCTION TO HISTORICAL NECESSITY

Modern modal logic began with necessity (or with things definable with respect to necessity), and the earliest literature, like C. I. Lewis [1918], confuses this with validity. Even in later work that is formally scrupulous about distinguishing these things, it is sometimes difficult to tell what concepts are really metalinguistic. Carnap, for instance, on p. 10 of Carnap [1956], begins his account of necessity by directing our attention to L-truth; a sentence of a semantical system (or language) is L-true when its truth follows from the semantical rules of the language, without auxiliary assumptions. This, of course, is a metalinguistic notion. But later, when he introduces necessity into the object language (Carnap [1956], p. 174), he stipulates that $\Box\varphi$ is true if and only if φ is L-true.

Carnap thinks of the languages with which he is working as fully determinate; in particular, their semantical rules are fixed. This has the consequence that whatever is L-true in a language is eternally L-true in that language. (See Schilpp [1963], p. 921, for one passage in which Carnap is explicit on the point: he says "analytic sentences cannot change their truth-value".) Combining this consequence with Carnap's explication of necessity, we see that[2]

(1) $\Box\varphi \rightarrow HG\Box\varphi$

will be valid in languages containing both necessity and tense operators: necessary truths will be eternally true. The combination of necessity with tense would then be trivialized.

But there are difficulties with Carnap's picture of necessity; indeed, it seems to be drastically misconceived.[3] For one thing, many things appear to be necessary, even though the sentences that express them can't be derived from semantical rules. In Kripke [1972], for instance, published 26 years after *Meaning and Necessity*, Saul Kripke argues that it is necessary that Hesperus is Phosphorous, though 'Hesperus' and 'Phosphorous' are by no means synonymous. Also at work in Kripke's conception of necessity, and that of many other contemporaries, is the distinction between φ expressing a necessary truth, and φ necessarily expressing a truth. In a well-known defense of the analytic-synthetic distinction, Grice and Strawson [1956] write as follows:

> Any form of words at one time held to express something true may, no doubt, at another time come to be held to express something false. But it is not only philosophers who would distinguish between the case where this happens as the result of a change of opinion solely as to matters of fact, and the case where this happens at least partly as a result of a shift in the sense of the words (p. 157).

This distinction, at least in theory, makes it possible that a sentence φ should necessarily (perhaps, because of semantical rules) express a truth, even though the truth that it expresses is contingent. This idea is developed most clearly in Kaplan [1978].

On this view of necessity, it attaches not primarily to sentences, but to propositions. A sentence will express a proposition, which may or may not be necessary. This can be explicated using possible worlds: propositions take on truth values in these worlds, and a proposition is necessary if and only if it is true in all possible worlds.[4]

This conception can be made temporal without trivializing the results. Probably the simplest way of managing this is to begin with nonempty sets T of times and W of worlds;[5] T is linearly ordered by a relation $<$. I will call this the $T \times W$ approach.

Recall that a tensed formula, say $F\varphi$, is true at $\langle w, t\rangle$, where $w \in W$ and $t \in T$, if and only if φ is true at $\langle w, t'\rangle$, for some t' such that $t < t'$.[6] We now want to ask under what conditions $\Box\varphi$ is true at $\langle w, t\rangle$. (In putting it this way we are suppressing propositions; this is legitimate, as long as we treat propositional attitudes as unanalyzed, and assume that sentences express the same proposition everywhere.)

If we appeal to intuitions about languages like English, it seems that we should treat formulas like $\Box\varphi$ as nontrivially tensed. This is shown most clearly by sentences involving the adjective 'possible', such as 'In 1932 it was possible for Great Britain to avoid war with Germany; but in 1937 it was impossible'. This suggests that when $\Box\varphi$ is evaluated at $\langle w, t\rangle$ we are considering what is *then necessary*; what is true in all worlds at that particular time, t.

The rule then is that $\Box\varphi$ is true at $\langle w, t\rangle$ if and only if φ is true at $\langle w', t\rangle$ for all $w' \in W$. If we like, we can make this relational. Let $\{\approx_t : t \in T\}$ be a family of equivalence relations on W, and let $\Box\varphi$ be true at $\langle t, w\rangle$ if and only if φ is true at $\langle t, w'\rangle$ for all $w' \in W$ such that $w \approx_t w'$.

The resulting theory generates some validities arising from the assumption that the worlds share a common temporal ordering. Formulas (2) and (3) are two such validities, corresponding to the principle that one world has a first moment if and only if all worlds do.

(2) $P[\varphi \vee \neg\varphi] \longleftrightarrow \Box P[\varphi \vee \neg\varphi]$

(3) $H[\varphi \wedge \neg\varphi] \longleftrightarrow \Box H[\varphi \wedge \neg\varphi]$

In case $\approx_t$ is the universal relation for every t (or the relations $\approx_t$ are simply omitted from the satisfaction conditions) there are other validities, such as (4) and (5).

(4) $P\Box\varphi \rightarrow \Box P\varphi$

(5) $F\Box\varphi \rightarrow \Box F\varphi$

As far as I know, the general problem of axiomatizing these logics has not been solved. But I'm not sure that it is worth doing, except as an exercise. The completeness proofs should not be difficult, using Gabbay's techniques (described in Section 4, below). And these logics do not seem particularly interesting from a philosophical point of view.

But a more interesting case is near to hand. The tendency we have noted to bring Carnap's metalinguistic notion of necessity down to earth has made room for the reintroduction of one of the most important notions of necessity: practical necessity, or historical necessity.[7] This is the sort of necessity that figures in Aristotle's discussion of the Sea Battle (*De Int.* $18^{b}25$–$19^{b}4$), and that arises when free will is debated. It also seems to be an important background notion in practical reasoning. Jonathan Edwards, in his usual lucid way, gives a very clear statement of the matter.

> Philosophical necessity is really nothing else than the full and fixed connection between the things signified by the subject and predicate of a proposition, which affirms something to be true.... [This connection] may be fixed and made certain, because the existence of that thing is already come to pass; and either now is, or has been; and so has as it were made sure of existence. And therefore, the proposition which affirms present and past existence of it, may be this means be made certain, and necessarily and unalterably true; the past event has fixed and decided that matter, as to its existence; and has made it impossible but that existence should be truly predicted of it. Thus the existence of whatever is already come to pass, is now become necessary; 'tis become impossible it should be otherwise than true, that such a thing has been (Edwards [1754], pp. 152–3).

Historical necessity can be fitted into the $T \times W$ framework; it is merely a matter of adjusting the relations $\approx_t$ so that if $w \approx_t w'$, then w and w' share the same past up to and including t. So for $t' < t$, atomic formulas must be treated the same way in w and w'. Furthermore, we have to stipulate that historical possibilities diminish monotonically with the passage of time: if $t < t'$, then $\{w' : w \approx_{t'} w'\} \subseteq \{w' : w \approx_t w'\}$. This interaction between time and relative necessity creates distinctive validities, such as (6) and (7).

(6) $\varphi \longleftrightarrow \Box\varphi$, if φ contains no occurrences of F

(7) $P\Box\varphi \rightarrow \Box P\varphi$

Formula (8), on the other hand, is clearly invalid.

(8) $\Box Pp \rightarrow P\Box p$

These correspond to rather natural intuitions relating the flow of time to the loss of possibilities.

There is another way of representing historical necessity, which perhaps will seem less straightforward to logicians steeped in possible worlds. Time can be treated as nonlinear (branching only towards the future), and worlds represented as branches on the resulting ordered structure. This corresponds very closely to the $T \times W$ account: (6) and (7) remain valid, and (8) invalid. But the validities are not the same. This matter will be taken up below, in Section 4.

So much for necessity; I will deal more briefly with 'ought' and conditionals.

As Aristotle points out, we don't deliberate about just anything; in particular, we deliberate only about what is in our power to determine. [*NE* 1112^{a}19f.] But the past, and the instantaneous present, are not in our power: deliberation is confined to future alternatives.

This suggests that deontic logic, insofar as it investigates practical oughts, should identify its possibilities with the ones of historical necessity. Unfortunately, this conception played little or no role in the early interpretation of deontic logic; those who developed the deontic applications of possible worlds semantics seemed to think of deontic possibilities ahistorically, as "perfect worlds" in which all norms are fulfilled.[8] Historical possibilities, on the other hand, are typically imperfect; life is full of occasions on which we have to make the best of a bad situation. In my opinion, this is one reason why deontic logic has seemed to most philosophers to consist largely of a sterile assortment of paradoxes, and why its influence on moral philosophy has been so fruitless.

Conditionals have been intensively studied by philosophical logicians over the last fifteen years, and this has created an extensive literature. Relatively little of this effort has been devoted to the interaction of conditionals with tense. But there is reason to think that important insights may be lost if conditionals are studied ahistorically. One very common sort of conditional (the philosopher's novel example of a "subjunctive conditional") is exemplified by (9).

(9) If Oswald hadn't shot Kennedy, then Kennedy would be alive today.

These conditionals seem to be closely related to historical possibilities; they envisage courses of events that diverge at some point in the past from the actual one. And this in turn suggests that there may be close connections between historical necessity and some conditionals.

Examples like the following four provide evidence of a different sort.

(10) He would go if she would go.
(11) He will go if she will go.
(12) He would have gone if she were to have gone.
(13) He went if she went.

Sentences (10) and (11) seem hardly to differ in meaning, if (10) has to do with the future. On the other hand, (12) and (13) are very different. If he didn't go, but would have gone if she had, (12) is true and (13) false.

This suggests that there may be systematic connections between tense, mood, and the truth conditions of conditionals. According to one extreme proposal, the difference in "mood" between (9) and the past-present conditionals form 'If Oswald didn't shoot Kennedy, then Kennedy is alive today' can be accounted for solely in terms of the interaction of tense operators and the conditional.[9] This has in favor of it the grammatical fact that 'would' is the past tense of 'will'. But the matter is complex, and it is difficult to see how much merit there is in the suggestion.

There have been recent signs of interest in the interaction of tense and conditionals; the most systematic of these is Thomason and Gupta [1981]. If this study is any indication, the topic is surprisingly complicated. But the complications may prove to be of philosophical interest.

The relation between historical necessity and quantum mechanics is a topic that I will not discuss at any great length. The indeterminacy that is associated with microphenomena seems at first glance to invite a treatment using alternative futures; and one of the approaches to the measurement problem in quantum theory, the "many-worlds interpretation", does appear to do just this. (See DeWitt and Graham [1973] for more information about the approach.)

But alternative futures don't provide in themselves an adequate representation of the physical situation, because the quantum mechanical probabilities can't be treated as distributions over a set of fully determinate worlds.[10] Some further apparatus would have to be introduced to secure the right sytem of nonboolean probabilities, and as far as I can see, what is required would have to go beyond the resources of possible worlds semantics: there is no escaping an analysis of measurement interactions, or of interactions in general.

Possible worlds semantics may help to make the "many worlds" approach to quantum indeterminacy seem less frothy; the prose of philosophical modal realists, such as D. Lewis [1970], is much more judicious than that in which the physicists sometimes indulge. (See, for instance, DeWitt [1973], p. 161.) So, modal logic may be of some help in sorting out the philosophical issues; but this leaves the fundamental problems untouched. Possible worlds are not in themselves a key to the problem of measurement in quantum mechanics.

The following sections will aim at fleshing out this general introduction with further historical information, more detailed descriptions of the relevant logical theories, and more extensive references to the literature.

3. HISTORICAL NECESSITY

The first sustained discussion of this topic, from the standpoint of modern tense logic is (as far as I know) Chapter 7 of Prior [1967], entitled 'Time and determinism'.[11] Prior's judgment and philosophical depth, as well as his readable style, make this required reading for anyone seriously interested in historical necessity.

Prior's exposition is informal, and sprinkled with historical references to the philosophical debate over determinism. In this debate he unearths a logical determinist argument, that probably goes back to ancient times. According to this argument, if φ is true then, at any previous time, $F\varphi$ must have been true. But choose such a time, and suppose that at this earlier time φ could have failed to come about; then $F\varphi$ could not have been true at this time. It seems to follow that the determinist principle

(14) $\varphi \rightarrow H\Box F\varphi$

holds good.

In discussing the argument and some ways of escaping from it, Prior is fairly flexible about this object language; in particular, he allows metric tense operators. Since these complicate matters from a semantic point of view, I will ignore them, and consider languages whose only modal operators are $\Box$ and the nonmetric tenses.

Also (and this is more unfortunate), Prior [1967] speaks loosely in describing models. At the place where his indeterminist models are introduced, for instance, he writes as follows.

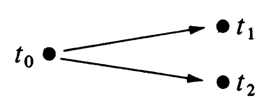
> ... we may define an Ockhamist *model* as a line without beginning or end which may break up into branches as it moves from left to right (i.e. from past to future), though not the other way ... (p. 126)

From this description it is clear that Prior is representing historical necessity by means of nonlinear time, rather than according to the $T \times W$ format described in Section 2, above. But it is a little difficult to tell exactly what mathematical structures have been characterized; probably, Prior had in mind trees whose branches all have the order type of the (negative and nonnegative) integers.

To bring this into accord with the usual treatment of linear nonmetric tense logic, we will liberalize Prior's account.

DEFINITION 1: A treelike frame $\mathfrak{A}$ for tense logic is a pair $\langle T, < \rangle$, where T is a nonempty set and $<$ is a transitive ordering on T such that if $t_1 < t$ and $t_2 < t$ then either $t_1 = t_2$ or $t_1 < t_2$ or $t_2 < t_1$.

As in ordinary tense logic, we imagine an assignment of truth values to atomic formulas at each $t \in T$, and truth-functional connectives are treated in the usual way. But things become perplexing when you try to interpret future tense in these structures. Take a very simple branching case, with just three moments, and imagine that p is true at t_0 and t_1, and false at t_2.



Is Fp true at t_0? It is hard to say.

Moreover, as you reflect on the problem, it becomes clear that Prior's juxtaposition of this technical problem with bits from figures like Diodorous Cronus, Peter de Rivo, and Jonathan Edwards is not merely an antiquarian quirk. There is a genuine connection. These treelike frames represent ways in which things can evolve indeterministically. A definition of satisfaction for a language with tense operators that is suited to such structures would automatically provide a way of making tense compatible with indeterministic cases. And it is just this that the logical argument for determinism claims can't be done. The technical problem can't be solved without getting to the bottom of this argument.

If the argument is correct, any definition of satisfaction for these structures will be incorrect – will generate validities that are at variance with the intended interpretation. Łukasiewicz's [1967] earlier three-valued solution is like this, I believe. Not because it makes some formulas neither true nor false, but because the formulas it endoreses as valid are so far off the mark. It is bad enough that $Fp \vee \neg Fp$ is invalid, but also the approach would make $[[\Diamond Fp \wedge \Diamond \neg Fp] \wedge [\Diamond Fq \wedge \Diamond \neg Fq]] \rightarrow [Fp \longleftrightarrow Fq]$ valid, if φ is true if and only if φ takes the intermediate truth value.[12]

Nor does the logic that Prior calls "Peircian" strike me as more satisfactory, from a philosophical standpoint, though it does lead to some interesting technical problems relating to axiomatizability. Here, $F\varphi$ is treated as true at t in case the moments at which φ is true bar the future paths through t; i.e., every branch through t contains a moment subsequent to t at which φ is true. On the Peircian approach, $F\varphi \vee \neg F\varphi$ is valid, but $Fp \vee F\neg p$ is not; nor is $p \rightarrow PFp$.[13] As Prior says, sense can be made of this by reading F as 'will inevitably'. Though this helps us to see what is going on, it is not the intended interpretation.

The most promising of Prior's suggestions for dealing with indeterminist future tense is the one he calls "Ockhamist". The theory will be easier to present if we first work out the satisfaction conditions for $\Box\varphi$ in treelike frames. Intuitively, $\Box\varphi$ is true at t if φ is true at t no matter what the future is like. And a way the future can be like will be represented by a fully determinate – i.e. linear – path beyond t. Since the frames are treelike, these correspond to the branches, or maximal chains, through t.

DEFINITION 2: Where $\langle T, <\rangle$ is a treelike model structure and $t \in T$, a branch through t is a maximal linearly ordered subset of T containing t; B_t is the set of branches containing t.

To make sense of φ being true at t no matter what the future is like, we will have to think of formulas being satisfied not just at moments t, but at pairs $\langle t, b\rangle$, where $b \in B_t$. Prior explains it this way. On the Ockhamist approach formulas like Fp are given '*prima facie* assignments' at t; such an assignment is made by choosing a particular b in B_t. His idea seems to be that there is something more provisional about the selection of b than about that of t; but this he does not articulate or defend very fully. In the technical formulation of the theory there is no asymmetry between moments and branches; it is just that two parameters need to be fixed in evaluating formulas.

Once satisfaction is made relative to pairs $\langle t, b\rangle$ for some formulas, it must be relativized in the same way for all formulas; otherwise the recursive definition of satisfaction will become snarled. So, except for future tense, the definition will go like this.

DEFINITION 3: A function h assigning each atomic formula a subset of T is called an (Ockhamist) *assignment*, as in Chapter II.2 of this *Handbook*.

DEFINITION 4: The h-truth value $\|\varphi\|^h_{\langle t,b\rangle}$ of φ at the pair $\langle t, b\rangle$ is defined as follows. We assume here that $\|\varphi\|^h_{\langle t,b\rangle} = 0$ iff $\|\varphi\|^h_{\langle t,b\rangle} \neq 1$.

$$\begin{array}{l}
\|\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad t \in h(\varphi), \quad \text{if } \varphi \text{ is atomic,}\\
\|\neg\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad \|\varphi\|^h_{\langle t,b\rangle} = 0,\\
\|\varphi \wedge \psi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad \|\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{and} \quad \|\psi\|^h_{\langle t,b\rangle} = 1,\\
\|\varphi \vee \psi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad \|\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{or} \quad \|\psi\|^h_{\langle t,b\rangle} = 1,\\
\|\varphi \rightarrow \psi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad \|\varphi\|^h_{\langle t,b\rangle} = 0 \quad \text{or} \quad \|\psi\|^h_{\langle t,b\rangle} = 1,\\
\|P\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad \text{for some } t' < t, \quad \|\varphi\|^h_{\langle t',b\rangle} = 1,\\
\|\Box\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff} \quad \text{for all } b' \in B_t, \quad \|\varphi\|^h_{\langle t,b'\rangle} = 1.
\end{array}$$

This definition renders $p \rightarrow \Box p$ valid, though not every substitution instance of it is valid. This can be easily changed by letting assignments take atomic formulas into subsets of $\{\langle t, b\rangle: t \in T \text{ and } b \in B_t\}$; Prior [1967], pp. 123–124, discusses the matter.

At this point, the way to handle $F\varphi$ is forced on us. We use the branch that is provided by the index.

$$\|F\varphi\|^h_{\langle t,b\rangle} = 1 \quad \text{iff for some } t' \in b \text{ such that } t < t', \|\varphi\|^h_{\langle t',b\rangle} = 1$$

The Ockhamist logic is conservative; it's easy to show that if φ contains no occurrences of $\Box$ then φ is valid for treelike frames if and only if φ is valid in ordinary tense logic. So indeterminist frames can be accommodated without sacrificing any orthodox validities. This is good for those who (like me) are not determinists, but feel that these validities are intuitively plausible. Finally, the Ockhamist solution thwarts the logical argument for determinism by denying that if φ is true at t (i.e. at $\langle b, t\rangle$, for some selected b in B_t) then $\Box\varphi$ is.

This way out of the argument bears down on its weakest joint; but the argument is so powerful that even this link resists the pressure; it is hard for an indeterminist to deny that $\Box\varphi$ must be true if φ is. To a thoroughgoing indeterminist, the choice of a branch b through t has to be entirely *prima facie*; there is no special branch that deserves to be called the "actual" future through t.[14] Consider two different branches, b_1 and b_2, through t, with $t < t_1 \in b_1$ and $t < t_2 \in b_2$. From the standpoint of t_1, b_1 is actual (at least, up to t_1). From the standpoint of t_2, b_2 is actual (at least, up to t_2). And neither standpoint is correct in any absolute sense. In exactly the same way, no particular moment of linear time is "present".

But then it seems that the Ockhamist theory gives no account of truth relative to a moment t, and it also suggests very strongly that if φ is true at t then $\Box\varphi$ is also true at t. The only way that a thing can be true at a moment is for it to be settled at that moment.

In Thomason [1970], it is suggested that such an absolute notion of truth can be introduced by superposing Van Fraassen's treatment of truth-value gaps onto Prior's Ockhamist theory.[15] The resulting definition is very simple.

DEFINITION 5:

$$\begin{array}{llll} \|\varphi\|_t^h = 1 & \text{iff} & \|\varphi\|_{\langle t,b\rangle}^h = 1 & \text{for all } b \in B_t, \\ \|\varphi\|_t^h = 0 & \text{iff} & \|\varphi\|_{\langle t,b\rangle}^h = 0 & \text{for all } b \in B_t. \end{array}$$

This logic preserves the validities of linear tense logic; indeed, φ is Ockhamist valid if and only if it is valid here. Also, the rule holds good that if $\|\varphi\|_t^h = 1$ then $\|\Box\varphi\|_t^h = 1$.

Thus, this theory endorses the principle (rejected by the Ockhamist theory) that if a thing is true at t then its truth at t is settled. It may seem at first that it validates all the principles needed for the logical determinist argument, but of course (since the logic allows branching frames) it must sever the argument somewhere. The way in which this is done is subtle. The scheme $\varphi \rightarrow HF\varphi$ is valid, but this does not mean that if φ is true at t then $F\varphi$ is true at any $t' < t$. That is, it is *not* the case that if $\|\varphi\|_t^h = 1$ then $\|F\varphi\|_{t'}^h = 1$ for all $t' < t$. The validity of this scheme only means that for all $b \in B_t$, if $\|\varphi\|_{\langle t,b\rangle}^h = 1$ then $\|F\varphi\|_{\langle t',b\rangle}^h = 1$ for all $t' < t$. But there may be $b' \in B_{t'}$ which are not in B_t.

To put it another way, the fact that $F\varphi$ is true at t from the perspective of *a later* t' does not make $F\varphi$ *absolutely* true at t, and so need not imply that $\Box F\varphi$ is true at t.

This maneuver makes use of the availability of truth-value gaps. To make this clearer, take a future-oriented version of the logical determinist argument:

$F\varphi \vee F\neg\varphi$ is true at any t; so $F\varphi$ is true at t or $\neg F\varphi$ is true at t; so $\Box F\varphi$ is true at t or $\Box\neg F\varphi$ is true at t. The supervaluational theory blocks the second step of this argument: in any such theory, the truth of $\psi \vee \neg\psi$ does not imply that ψ is true or $\neg\psi$ is true.

Thomason suggests that this logic represents the position endorsed by Aristotle in *De Int.* $18^{b}25$-$19^{b}4$, but his suggestion is made without any analysis of the very controversial text, or discussion of the exegetical literature. For a close examination of the texts, with illuminating philosophical discussion, see Frede [1970]; also see Sorabji [1980]. For a broadly-based examination of Aristotle's views that nicely illustrates the value of treelike frames as an interpretive device, see Code [1976]. The suggestion is also made in Jeffrey [1979]. For information about the medieval debate on this topic, see Normore [1982].

4. THE TECHNICAL SIDE OF HISTORICAL NECESSITY

The mathematical dimension of the picture painted in the above section is still relatively undeveloped. At present, most of the results known to me deal with axiomatizability in the propositional case.

We have already characterized one important variety of propositional validity: φ is *Ockhamist valid* if it is satisfied at all pairs $\langle t, b\rangle$, relative to all Ockhamist assignments on all treelike frames. (See Definitions 2–4, above.) The time has come to give an official defintion of $T \times W$ validity.

DEFINITION 6: A $T \times W$ frame is a quadruple $\langle W, T, <, \approx\rangle$, where W and T are nonempty sets, $<$ is a transitive relation on T which is also irreflexive and linear (i.e. $t \not< t$ for all $t \in T$, and either $t < t'$ or $t' < t$ or $t = t'$ for all, t, $t' \in T$), and $\approx$ is a 3-place relation on $T \times W \times W$, such that (1) for all t, $\approx_t$ is an equivalence relation (i.e. $w \approx_t w$ for all $t \in T$ and $w \in W$, etc.), and (2) for all w_1, $w_2 \in W$ and t, $t' \in T$, if $w_1 \approx_t w_2$ and $t' < t$ then $w_1 \approx_t w_2$. The intention is that $w \approx_t w'$ if w and w' are historical alternatives through t, and so differ only in what is future to t.

DEFINITION 7: A function h assigning each atomic formula a subset of $T \times W$ is an *assignment*, provided that if $w \approx_t w'$ and $t_1 \leqslant t$ then $\langle t_1, w\rangle \in h(\varphi)$ iff $\langle t_1, w'\rangle \in h(\varphi)$.

DEFINITION 8: The h-truth value $\|\varphi\|^{h}_{\langle t, w\rangle}$ of φ at the pair $\langle t, w\rangle$ is defined by a recursion that treats truth-functional connectives in the usual way. The clauses for tense and necessity run as follows.

$$\begin{aligned}
\|P\varphi\|^h_{\langle t,w\rangle} &= 1 \quad \text{iff for some } t' < t, \quad \|\varphi\|^h_{\langle t',w\rangle} = 1,\\
\|F\varphi\|^h_{\langle t,w\rangle} &= 1 \quad \text{iff for some } t' \text{ such that } t < t', \quad \|\varphi\|^h_{\langle t',w\rangle} = 1,\\
\|\Box\varphi\|^h_{\langle t,w\rangle} &= 1 \quad \text{iff for all } w' \text{ such that } w \approx_t w', \quad \|\varphi\|^h_{\langle t,w'\rangle} = 1.
\end{aligned}$$

A formula is $T \times W$ valid if it is satisfied at every pair $\langle t, w\rangle$ by every assignment on every $T \times W$ frame.[16]

There are some validities that are peculiar to these $T \times W$ frames, and that arise from the fact that only a single temporal ordering is involved in these frames: (15) and (16) are examples,

(15) $FG[p \wedge \neg p] \rightarrow \Box FG[p \wedge \neg p]$

(16) $GF[p \vee \neg p] \rightarrow \Box GF[p \vee \neg p]$

Example (15) is valid because its antecedent is true at $\langle t, w\rangle$ if an only if there is a t' that is $<$-maximal with respect to w; but this holds if and only if there is a t' that is $<$-maximal absolutely. Example (16) is similar, except that this time what is at stake is the nonexistence of a maximal time.

Burgess remarked (in correspondence) that $T \times W$ validity is recursively axiomatizable, since it is essentially first-order. But as far as I know the problem of finding a reasonable axiomatization for $T \times W$ validity is open. I would expect the techniques discussed below, in connection with Kamp validity, to yield such an axiomatization.

Although (15) and (16) may be reasonable given certain physical assumptions, they do not seem so plausible from a logical perspective. After all, if $w \approx_t w'$, all that is required is that w and w' should share a certain segment of the past, and this implies that the structure of time should be the same in w and w' on this segment. But it is not so clear that w and w' should participate in the same temporal structure after t. Thus suggests a more liberal sort of $T \times W$ frame, first characterized by Kamp [1979].[17]

DEFINITION 9: A Kamp frame is a triple $\langle \mathscr{T}, W, \approx\rangle$, where W is a nonempty set, $\mathscr{T}$ is a function from W to transitive, irreflexive linear orderings (i.e. if $w \in W$ then $\mathscr{T}(w) = \langle T_w, <_w\rangle$, where $<_w$ is an ordering on T_w as in $T \times W$ frames), and $\approx$ is a relation on $\{\langle t, w, w'\rangle : w, w' \in W \text{ and } t \in \mathscr{T}(w) \cap \mathscr{T}(w')\}$ such that for all t, $\approx_t$ is an equivalence relation, and if $w \approx_t w'$ then $\{t_1 : t_1 \in T_w \text{ and } t_1 <_w t\} = \{t_1 : t_1 \in T_{w'} \text{ and } t_1 <_{w'} t\}$. Also, if $w \approx_t w'$ and $t' <_w t$ then $w \approx_{t'} w'$.

The definitions of an assignment, and of the h-truth value $\|\varphi\|^h_{\langle t,w\rangle}$ of φ at the pair $\langle t, w\rangle$ (where $t \in \mathscr{T}_w$ and h is an assignment) are readily adapted from Definitions 7 and 8.

Besides (AK0) all classical tautologies, Kamp takes as axioms all instances of the following schemes.[18]

(AK1) $H[\varphi \rightarrow \psi] \rightarrow [P\varphi \rightarrow P\psi]$
(AK2) $G[\varphi \rightarrow \psi] \rightarrow [F\varphi \rightarrow F\psi]$
(AK3) $PP\varphi \rightarrow P\varphi$
(AK4) $FF\varphi \rightarrow F\varphi$
(AK5) $PG\varphi \rightarrow \varphi$
(AK6) $FH\varphi \rightarrow \varphi$
(AK7) $PF\varphi \rightarrow [P\varphi \vee \varphi \vee F\varphi]$
(AK8) $FP\varphi \rightarrow [P\varphi \vee \varphi \vee F\varphi]$
(AK9) $\Box[\varphi \rightarrow \psi] \rightarrow [\Box\varphi \rightarrow \Box\psi]$
(AK10) $\Box\varphi \rightarrow \varphi$
(AK11) $\Diamond\varphi \rightarrow \Box\Diamond\varphi$
(AK12) $\Diamond P\varphi \rightarrow P\Diamond\varphi$
(AK13) $\Box\varphi \vee \Box\neg\varphi$, if φ is atomic

As well as (RK0) *modus ponens*, Kamp posits the rules given by the following three schemes.

(RK1) $\varphi/H\varphi$
(RK2) $\varphi/G\varphi$
(RK3) $\varphi/\Box\varphi$

Readers familiar with axioms for modal and tense logic will see that this list falls into three natural parts. Classical tautologies, *modus ponens*, (RK1), (RK2) and (AK1)–(AK8) are familiar principles of ordinary tense logic without modality. Classical tautologies, *modus ponens*, (RK3), and (AK9)–(AK11) are principles of the modal logic **S5**. Axiom (AK12) is a principle combining tense and modality; this principle was explained informally in Section 2. The validity of (AK13) reflects the treatment of atomic formulas as noncontingent; see the provision in Definition 7. If tense operators were not present, (AK13) would of course trivialize $\Box$, rendering every formula noncontingent.

To establish the incompleteness of (AK0)–(AK13) + (RK0)–(RK3), consider the formula (17), discovered by Kamp, where $E_1(\varphi)$ is $F\varphi \wedge G[\varphi \vee P\varphi]$.

(17) $[PE_1(p) \wedge \Diamond PE_1(q)] \rightarrow [P[E_1(p) \wedge P\Diamond E_1(q)] \vee$
$\vee P[\Diamond E_1(q) \wedge \Diamond E_1(p)] \vee P[E_1(p) \wedge \Diamond E_1(q)]]$.

The validity of (17) in Kamp frames follows from the fact that these frames are closed under the sort of diagram completion in given in Figure 1. Given t_1, t_2, t_3, t_1', t_3' and the relations of the diagram, it must be possible to

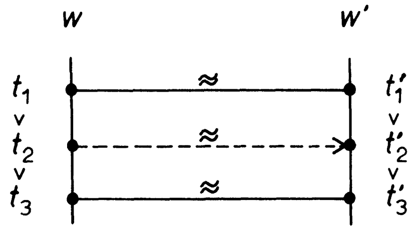


Fig. 1. Interpolation.

interpolate at t'_2 in w' alternative to t_2, with $t'_3 < t'_2 < t'_1$.[19] Formula (17) is complicated, but I think I can safely leave the task to checking its Kamp validity to the reader.

One proof that (17) is independent of (AK0)–(AK13) + (RK0)–(RK3) makes use of still another sort of frame, which is closer to a Henkin construction than the $T \times W$ frames. If our task is to build models out of maximal consistent sets of formulas, the 'times' of a $T \times W$ frame are rather artificial; they would have to be equivalence sets of maximal consistent sets. In *neutral frames*, the basic elements are moments, or instantaneous slices of evolving worlds, which are organized by intra-world temporal relations, and interworld alternativeness relations. Neutral frames tend to proliferate, because of the many conditions that can be imposed on the relations; hence my use of subscripts in describing them.

DEFINITION 10: A neutral frame$_1$ is a triple $\langle W, \mathscr{U}, \approx \rangle$, where (1) W is a nonempty set, (2) $\mathscr{U}$ is a function whose arguments are members w of W and whose values are orderings $\langle U_w, <_w \rangle$ such that U_w is a nonempty set and $<_w$ is a transitive[20] ordering on U_w such that for all $a, b \in U_w$ either $a < b$ or $b < a$ or $a = b$, (3) if $w, w' \in W$ and $w \neq w'$ then $\mathscr{U}_w$ and $\mathscr{U}_{w'}$ are disjoint, and (4) $\approx$ is an equivalence relation on $\cup \{U_w : w \in W\}$, and (5) if $a \approx a'$ and $b <_w a$ then there is some $b' \in \mathscr{U}_{w'}$ (where $a' \in \mathscr{U}_{w'}$) such that $b \approx b'$ and $b' <_w a'$.

Here, each U_w corresponds to the set of instantaneous slices of the world w; $\approx$ is the alternativeness relation between these slices.

The diagram-completion property expressed in (5) looks as shown in this picture. It's important to realize that nothing prevents $a \neq b$ while at the same time $a' = b'$ in Figure 2. Of course, in this case we will also have $a \approx b$, which would be something like history repeating itself, at least in all respects that are settled by the past.

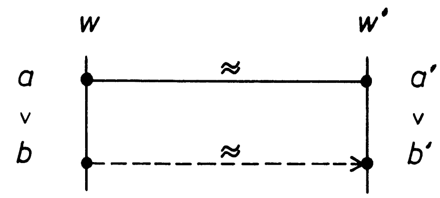


Fig. 2. One-way completion.

Everything generated by (AK0)–(AK13) + (RK0)–(RK3) is valid in neutral frames$_1$. (And in fact the converse holds, so that we have a completeness result; see Thomason [1981c].) But it is easy to show that (17) is invalid in neutral frames$_1$.

This process can be continued; for instance, stronger conditions of diagram completion can be imposed on neutral frames, which extend the propositional validities, and completeness conditions obtained for the resulting sorts of frames. Details can be found in Thomason [1981c]; but this effort did not produce an axiomatization of Kamp validity.

Using his method of constructing irreflexive models (see Gabbay [1981]), Gabbay has shown that all the Kamp validities will be obtained if (AG1) and (RG1) are added to (AK0)–(AK13) + (RK0)–(RK3). Some terminology is needed to formulate (RG1); we need a way of talking, to put it intuitively, about formulas which record a finite number of steps forwards, backwards, and sideways in Kamp frames.

DEFINITION 11: Let $\alpha_i \in \{P, F, \Diamond\}$ for $1 \leqslant i \leqslant n, n \geqslant 0$. Then $f(\varphi) = \varphi$, and

$$f_{\alpha_1,\ldots,\alpha_n}(\varphi_0, \ldots, \varphi_{n-1}, \varphi_n) \\ = \varphi_0 \wedge \alpha_1[\varphi_1 \wedge \ldots \wedge \alpha_{n-1}[\varphi_{n-1} \wedge \alpha_n\varphi_n] \ldots].$$

So, for instance, $f_{F,\Diamond,P}(p_0, p_1, p_2, p_3)$ is $p_0 \wedge F[p_1 \wedge \Diamond[p_2 \wedge Pp_3]]$.

The axiom and rule are as follows:

(AG1) $[\Box\neg\varphi \wedge H\Box\varphi \wedge \Box\psi] \rightarrow G\Box H[[\neg\varphi \wedge H\varphi] \rightarrow \psi]$

(RG2) $$\frac{f_{\alpha_1,\ldots,\alpha_n}(\varphi_0, \ldots, \varphi_{n-1}, [\varphi_n \wedge \neg p \wedge Hp]) \rightarrow \psi}{f_{\alpha_1,\ldots,\alpha_n}(\varphi_0, \ldots, \varphi_{n-1}, \varphi_n) \rightarrow \psi}$$

In RG2, p must be foreign to ψ and $\varphi_0, \ldots, \varphi_n$.

I leave it to the reader to verify that the axiom and rules are Kamp valid.

It looks as if the ordering properties of linear frames that can be axiomatized in ordinary tense logic (endlessness towards the past, endlessness towards the future, density, etc.)[21] can be axiomatized against the background of Kamp frames, using Gabbay's techniques. But even so, there are still many simple questions that need to be settled; to take just two examples, it would be nice to know whether Kamp satisfaction is compact, and whether (RG2) is independent.

The situation with respect to treelike frames (i.e. with respect to Ockhamist validity) was even less well explored until recently. To begin with, a number of people have noticed that, although every propositional formula that is Kamp valid is Ockhamist valid,[22] the converse fails. Nishimura [1979b] points this out, giving (18) as a counterexample.[23]

(18) $GH\Box FP[H\neg p \wedge \neg p \wedge Gp] \rightarrow FP\Diamond FP[\neg p \wedge \Box Gp]$

In 1977, Burgess discovered (19), the simplest counterexample known to me.

(19) $\Box G\Diamond Fp \rightarrow \Diamond GFp.$

And in 1978, Thomason independently constructed the following counterexample.

(20) $[p \wedge \Box GH[p \rightarrow Fp]] \rightarrow GFp$

Both examples trade on the fact that any linearly-ordered subset of a tree can be extended to a branch (a consequence of the Axiom of Choice). If the antecedent of (20) is true at $\langle t, b\rangle$ in a frame $\langle T, <\rangle$ then there is a linear set X of moments such that p is true at every member of X, and such that there is no upper bound to T to X. This set X can be extended to a branch b^*, and GFp will be true at $\langle t, b^*\rangle$.

But the situation shown in Figure 3 can arise in Kamp models, allowing (19) to be falsified. If we make a Kamp model out of $\{b_i : i \in \omega\}$, omitting b_ω, (19) is false at $\langle t_0, b_0\rangle$.

Nishimura [1979a, b] seems to feel that these examples show treelike frames to be inadquate. But the technical results only establish a difference between the treelike and the $T \times W$ approaches. Adequacy has to do with intuitions about what should be valid. Intuitions may differ, but to me the natural notion is that of a possible future – not that of a possible course of events. Thus, (20) strikes me as clearly valid. The metric examples discussed in Nishimura [1979a] affect me similarly. The person who holds both (21) and (22) seems to me to have contradicted himself.

(21) Inevitably, life on earth will come to an end at some date in the future.

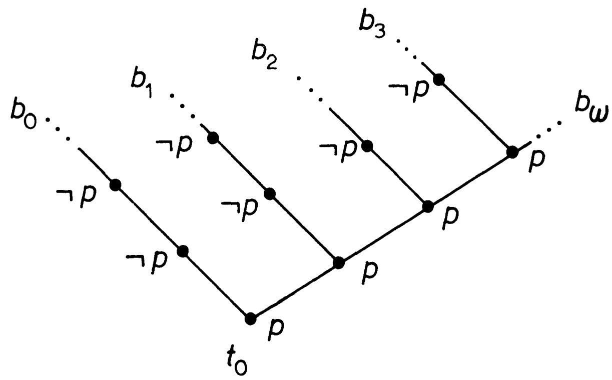


Fig. 3.

(22) For every date in the future, it is not inevitable that life on earth will have come to an end by that date.

For this reason, it seems to me that the $T \times W$ frames do not have the philosophical interest of the treelike ones, though they are certainly interesting for technical reasons. This makes Ockhamist validity appear worth investigating; until recently, however, very little was known about it.[24] In Burgess [1979], it is claimed that Ockhamist validity is recursively axiomatizable, and a proof is sketched. Later (in conversation), Kripke challenged the proof, and Burgess has been unable to substantiate all the details. Very recently, Gurevich and Shelah have proved a result implying that Ockhamist validity is decidable. (See Gurevich and Shelah [to appear].) At present (October, 1982) their paper is not yet written, and I have not had an opportunity to see their proof. The main result is that the theory of trees with second-order quantification over maximal chains is decidable.

Of course, a proof of decidability would allow axioms to be recovered for Ockhamist validity; but this would be done in Craigian fashion. And unfortunately, Gabbay's completeness techniques do not seem (at first glance, anyway) to extend to the treelike case. Burgess' example [1979], p. 577, of an Ockhamist invalid formula valid in countable treelike frames, helps to bring home the complexity of this case.[25]

There are some interesting technical results regarding logics other than the treelike and $T \times W$ ones that I have stressed here; the most important of these is Burgess' [1980] proof that the Peircian validities are decidable. Burgess has pointed out to me that the method of Section 5 of Burgess [1980] can be used to prove Kamp validity decidable, as well as $T \times W$

validity with dense time. He also remarks that most interesting technical questions about the case in which atomic formulas are treated like complex ones (so that $p \rightarrow \Box p$ is not valid, and substitution is an admissible rule) are unresolved, and may prove more difficult.

5. DEONTIC LOGIC COMBINED WITH HISTORICAL NECESSITY

For general discussion of deontic logic, with historical background, see Føllesdal and Hilpinen [1971] and Chapter II.12 of this *Handbook*. This presentation will concentrate on combinations of deontic modalities with temporal ones, and, in particular, with historical necessity.

Deontic logic seems to have suffered from a lack of communication. Even now, papers are written in which the relevant literature is not mentioned and the authors appear to be reinventing the wheel. In the hope that a survey of the literature will help to correct this situation, I have tried to make the bibliographical coverage of the present discussion thorough.

One facet of this lack of communication can be seen at work in the late 1960s. On the one hand, quite sophisticated model theoretic studies were developed during this time, treating deontic possibilities historically, as future alternatives. Montague [1968], pp. 116–117 and Scott [1967] represent the earliest such studies. (Unfortunately, Montague's presentation is tucked away in a rather forbidding technical paper that discusses many other topics, and the publication of the paper was delayed. And Scott's paper was never published.) But in 1969 Chellas [1969] appeared, giving an extended and very readable presentation of the California Theory.[26] (Montague's, Scott's, and Chellas' theories are quite similar variations of the $T \times W$ approach; the treatment of historical necessity is similar, and indeed identical in all important respects, to Kamp's.)

But although Chellas' monograph contains an extensive and valuable bibliography of deontic logic, including many references to the literature in moral philosophy and practical reasoning, and though Chellas is evidently familiar with this literature, there is no attempt in the work to relate the theory to the more general philosophical issues, or even to discuss its application to the "paradoxes" of deontic logic, which by then were well known Chellas concentrates on the mathematical portion of the task. Thus, although the presentation is less compressed than Montague's it remains relatively impenetrable to most moral philosophers, and there is no advertisement of the genuine help that the theory can give in dealing with these puzzles.

On the other hand, in the philosophical literature it is easy to find studies that would have benefited from vigorous contact with logical theories such as Chellas'. To consider an example almost at random, take Chisholm [1974]. This paper has the word 'logic' in its title and deals with a topic that is thoroughly entangled with Chellas' investigation, but Chisholm's paper has no references to such logical work and seems entirely ignorant of it. Moreover, the paper is written in an axiomatic style that makes no use of semantical techniques that had been current in the logical literature for many years. And Chisholm commits errors that could have been avoided by awareness of these things. (D9 on p. 13 of Chisholm [1974] is an example; compare this with pp. 183–184 of Thomason [1981b].)

To take another example, Wiggins [1973] provides an informal discussion, within the context of the determinist-libertarian debate in moral philosophy, of issues very similar to those treated by Chellas [1971] (and, so far as historical necessity goes, in Chapter 7 of Prior [1967]). Again, the paper contains no references to the logical literature. Though Wiggins' firm intuitive grasp of the issues prevents his argument from being affected,[27] it would have been nice to see him connect his account to the very relevant model theoretic work.

There are a number of general discussions of the "paradoxes" of deontic logic: see, for example, Åqvist [1967], pp. 364–373, Føllesdal and Hilpinen [1971], pp. 21–26, Hannson [1971], pp. 130–133, Al-Hibri [1978], pp. 22–29, and Van Eck [1981], pp. 28–35. It should be apparent from the shudder quotes that I prefer not to dignify these puzzles with the same term that is applied to profoundly deep (perhaps unanswerable) questions like the Liar Paradox. The puzzles are a disparate assortment, and require a spectrum of solutions, some of which have little to do with tense. The Good Samaritan problem, for instance (the problem of reparational obligations), seems from one point of view to simply be a rediscovery of the frailties of the material conditional as a formalization of natural language conditionals.

But part of the solution[28] of this problem seems to lie in the development of an *ought kinematics*,[29] in analogy to the probability kinematics that is the topic of Jeffrey [1965], Chapter 11. As we would expect from probability, where there are interactions (surprisingly complex ones) between the rules of probability kinematics and locutions that combine conditionality and probability, we should expect there to be close relationships between ought kinematics and the semantics of conditional oughts. Nevertheless, we can formulate the kinematics of ought without having to work with conditional oughts in the object language.

The technical resolution of this problem is very simple, and this is precisely what the California theory that I referred to above provides: e.g., the theory of Chellas [1969]. For the sake of variety, I will formulate it with respect to treelike frames; this is something that, to my knowledge, was first done in Thomason [1981a].[30]

DEFINITION 12: A treelike frame $\mathfrak{A}$ for deliberative deontic tense logic is a pair $\langle T, <, O\rangle$, where $\langle T, <\rangle$ is a treelike frame for tense logic and O is a function on T such that for all $t \in T$, (1) O_t is nonempty, (2) $O_t \subseteq B_t$, and (3) if $t < t'$ and $t' \in b$ for some $b \in O_t$, then $O_{t'} = O_t \cap B_{t'}$.

The satisfaction clause for oughts is as follows.

$$\|O\varphi\|^{h}_{\langle t,b\rangle} = 1 \quad \text{iff for all } b' \in O_t, \quad \|\varphi\|^{h}_{\langle t,b'\rangle} = 1.$$

We are dealing with the result of extending a propositional language of the sort we discussed in previous sections (truth-functional connectives, past and future tense, and historical $\Box$) by adding a one-place connective O for ought.[31]

Something needs to be said about Condition (3) on frames, which is not to be found in Thomason [1981a], and, as far as I know, has also been overlooked by other authors who (like Chellas) formulate the model theory of ought kinematics. This condition yields validities such as the following two.

(23) $OG[F\varphi \rightarrow \neg OG\neg\varphi]$

(24) $OG\varphi \rightarrow OGO\varphi$

These do strike me as valid in this context, and at any rate are interesting tense-deontic principles having to do with the coherence of plans. Principle (23), for instance, disallows an alternative future in O_t along which some outcome will happen, but is forbidden from ever happening. Such an alternative can't correspond to a coherent plan.

This setup can readily deal with reparational obligations. Suppose that, because of a promise to my aunt, at 4:00 I ought to catch an airplane at 5:00, but that at 5:00 I have broken my promise because of the attractions of the airport bar. Then at 5:00 I should call my aunt to tell her I won't be on the plane. Though this is the sort of situation that is sometimes represented as paradoxical in the literature, it is easily modeled in ought kinematics, with no apparent conceptual strain. At one time we have $O\neg Fp$ true, where p stands for 'I tell my aunt I won't be on the plane'. At a later time (one that involves the occurrence of something that shouldn't have happened) OFp is true.

If we press the account a bit harder, we can change the example; it also is true at 5:00 that I ought to inform my aunt I won't be on the plane, and this can be taken to entail that I ought not to be on the plane, since I can only inform someone of what is true, and because $\Box[\varphi \to \psi] \to [O\varphi \to O\psi]$ is a validity of our logic.

The response to this pressure is, of course, a Gricean maneuver;[32] it is true at 5:00 (on the understanding of 'ought' in question) that I ought not to be on the plane. But it is not worth saying at 5:00, and if I were to say it then, I would be taken to have said something else, something false. And all of this can be made plausible in terms of general principles of reasonable conversation. I will not give the details here, since they can easily be reconstructed from D. Lewis [1979b].[33] On balance, the approach seems to be well braced against pressure from this direction. It is hard to imagine a reasonable theory of truth conditions that will not have to deploy Gricean tactics at some point. And this can be made to look like a very reasonable place to deploy them.

These linguistic reflections are nicely supported by quite independent philosophical considerations. Greenspan [1975] is a sustained study of ought kinematics from a philosophical standpoint, in which it is argued that a time-bound treatment of oughts is essential to an understanding of their logic, and that the proper view of deontic detachment is that a conditional ought licenses a "consequent ought" when the antecedent is unalterably true. The paper contains many useful references to the philosophical literature, and provides a good example of the results that can be obtained by combining this philosophical material with the logical apparatus.

The fact that 'I ought not to be on the plane' would ordinarily be taken to be false at 5:00 in the example we discussed above shows that 'ought' has employments that are not practical or deliberative: ones that perhaps have to do with wishful thinking. Also, there is its common use to express a kind of necessity: 'The butter is warm enough now; it ought to melt'.

Philosophers are inclined to speak of ambiguity in cases like this, but this is either a failure to appreciate the facts or an abuse of the word 'ambiguity'. The word 'ought' is indexical or context-sensitive, not ambiguous. The matter is argued, and some of its consequences are explored, in Kratzer [1977]; see D. Lewis [1979b] for a study of the consequences in a more general setting.

In Thomason [1981a] this is taken into account by a more general interpretation of O, according to which the relevant alternatives at t need not be possible futures for t. Probably the most general account would make the interpretation of O in a context relative to a set of alternatives which are regarded as possible in some sense relative to that context.

This is of course related to the philosophical debate over whether 'ought' implies 'can'.[34] The most sophisticated linguistic account makes the issue appear rather boring: if we attend to a reasonable distinction between ambiguity and context-sensitivity, 'ought' doesn't imply 'can', since there are contexts that provide counterexamples. But in practical contexts, when the one is true the other will be, even if for technical reasons we can't relate this to an implication among linguistic forms.[35] This result is rather disappointing. But maybe there are ways of extracting interesting consequences for moral philosophy from a pragmatic account of oughts and other practical phenomena. An idea that I find intriguing is that manipulation of the context is the typical – perhaps the only – mechanism of moral weakness. The idea is suggested in Thomason [1981b], but is not much developed.

There seems to be no point in discussing the technical side of temporal deontic logic here. Not much work has been done in the area,[36] and the best strategy seems to be to let the matter wait until more is known about the interpretation of historical necessity.

Strictly speaking, conditional oughts are more closely related to the combination of conditionals with oughts than that of tense with modalities. Because the topic is complex and a thorough discussion of it would take up much space I have decided to neglect it in this article, even though (as Greenspan [1975] makes clear) tense enters into the matter. For an illuminating discussion of some of the problems, see DeCew [1981].[37]

6. CONDITIONAL LOGIC COMBINED WITH HISTORICAL NECESSITY

In the present essay, 'modality' has been confined to what can be interpreted using possible worlds semantics. So here, 'conditional logic' has to do with the modern theories that were introduced by Stalnaker and then by D. Lewis; see Stalnaker [1968] and D. Lewis [1973].[38] For surveys of this work, see Chapter II.9 of this *Handbook*, and Harper [1981] and the volume in which it appears: Harper *et al.* [1981].

The interaction of conditionals and historical necessity is a topic that is only beginning to receive attention.[39] As in the case of deontic logic there is a fairly venerable philosophical tradition, involving issues that are still debated in the philosophical literature, and a certain amount of technical model theoretic work that may be relevant to these issues. But this time, the philosophical topic is causality.[40]

A conditional like D. Lewis', which does not satisfy the principle of